

Make Your Own UV Sterilisation Box

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Due to coronavirus we are now used to alcohol based sanitisation liquids and sprays but what about disinfecting items like goggles, keys, mobile phone, watch, currency notes, laptop, courier packets, etc that we handle? A convenient solution for disinfecting such items is an ultraviolet (UV) sterilisation box that you can build yourself.

Ultraviolet is electromagnetic radiation with wavelength of 10 to 400nm. It has shorter wavelength than visible light but longer than x-rays. The wide range of UV radiation is grouped into three main categories:

- UV-A with wavelength of 315 to 400nm
- UV-B with wavelength of 280 to 315nm
- UV-C with wavelength of 100 to 280nm

Across the world, UV-C sterilisation is an established method for inactivating pathogens, viruses, and bacteria. UV-C disrupts the DNA of pathogens and impairs their ability to perform vital cellular functions. The effectiveness of this germicidal exposure depends on the duration of exposure and the intensity and wavelength

of the UV radiation. It is therefore necessary that sufficient UV-C light reaches all parts of the object being disinfected.

Germicidal UV-C is commonly generated using low-pressure mercury vapour lamps, which emit light at 253.7nm. UV-C LEDs that generate light at wavelengths between 255 and 280nm are also available. In the prototype, we have used two 11-watt germicidal UV tubes along with their chokes.

Features of this steriliser are:

- UV-C lamps operating at 254nm wavelength for effective germicidal effect
- Enclosed UV leak-proof box
- Reflective sides for maximising reflections and giving effective 360-degree coverage
- Magnetic hold of door/lid to keep it securely closed
- Rotary switch for setting exposure time from 1 to 10 minutes
- Timer can be paused, if desired
- Simple operation using Start/Resume and Stop/Pause buttons
- Display showing remaining time and alert messages
- Safety feature that shuts off the

UV lamp if the door/lid of the box is opened

Construction

The UV steriliser can be made using a suitable-size stiff cardboard or wooden box, or any other sturdy box. (Direct exposure to UV-C rays is harmful to humans, so build a proper enclosure.) The author's prototype was built using an empty printer box as shown in Fig. 1.

The top of the box was sealed and fixed while a slot was cut on the front side of the box as shown in Fig. 2. A 3cm edge (frame) was left on all sides of the opening so that the door could rest (close) along these edges. The door can be built using a cardboard sheet that is at least 2cm bigger than the opening to block the UV light properly.

A hinge can be made on one edge by cutting a 3cm strip from the door and sticking it again with a sturdy tape or a cloth strip. On the opposite edge three neodymium magnets can be embedded inside the cardboard to serve as a secure lock for the door. A thin iron strip (such as a piece of an old hacksaw) may be attached to



Fig. 1: Author's prototype



Fig. 2: Opening on the front side